Domination Parameters and the Removal of Matchings

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Abstract

We consider the effect of the removal of a matching from a graph on the value of domination and domination-related parameters with emphasis on the removal of a maximum matching from a cubic graph or tree. For example, we show that the independent domination number of a nontrivial tree increases on the removal of any maximum matching.

1 Introduction

For the domination number, like many other graph parameters, the effect of adding or removing vertices or edges is well-studied. For example, the minimum number of edges whose removal increases the domination number is known as the bondage number [7], and there are versions for other domination parameters such as independent bondage [12]. (See also for example [10, 13] and the references therein.) Recently, we [4] introduced the idea of removing a matching. That paper focused on trees and considered only the domination number. Here we consider the problem in a more general setting. Specifically we consider the effect on some domination parameters of removing a matching or a maximum matching from a graph.

We use $\gamma(A)$ to denote the domination number, $i(A)$ to denote the independent domination number, $\gamma_t(A)$ to denote the total domination number, and $\alpha(A)$ to denote the independence number of a graph A. We posit throughout that A is connected (or at least isolate-free).

We proceed as follows. In Section 2 we provide some bounds for general graphs. In Section 3 we focus on trees, and show, for example, that while there are

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trees T and matchings M such that $i(T-M) < i(T)$, for any maximum matching M_m it holds that $i(T-M_m) > i(T)$ (unless T is K_1). Thereafter we focus on cubic graphs Q of order n. In Section 4 we show that $\gamma(Q-M_m) \geq 10n/33$ for any maximum matching M_m, and there exist cubic graphs and maximum matchings such that $\gamma(Q-M_m) = 4n/13$. In Sections 5 and 6 we show for example, that while there are cubic graphs Q and matchings M such that $\alpha(Q-M) > n/2$ and $\gamma_t(Q-M) < n/2$, for any maximum matching M_m it holds that $\alpha(Q-M_m) \leq n/2 \leq \gamma_t(Q-M_m)$.

2 General Bounds

We begin by considering, for ψ one of our parameters, upper and lower bounds on $\psi(G-M)$ that hold for all graphs G of a particular order n and maximum matchings M thereof. Lower bounds are trivial and achieved by the complete graph; the bound is 1 for γ and i when n is odd; and 2 in all other cases (that is, γ and i when n is even, and γ_t and α for all n).

We consider upper bounds next. For domination, in [4] we showed that $\gamma(G-M) \leq 3n/4$, and that this is achieved by the corona of a corona (or the corona of C_4) with M a perfect matching. (Recall that the corona of a graph H is obtained by adding for each vertex v of H one end-vertex adjacent only to v; see Figure 3 for an example where H is a star.) For independent domination, Favaron [6] showed that $i(G) \leq n + 2 - 2\sqrt{n}$ for a graph without isolates. From this we obtain:

Lemma 1 *If G is a connected graph on n vertices and M a matching of G, then $i(G-M) \leq n + 2 - \sqrt{2n}$, and this bound is sharp.*

Proof. The result is immediate if $n \leq 2$; so assume $n \geq 3$. Say $G - M$ has x isolated vertices. Then by Favaron's bound, it holds that $i(G-M) \leq x + \left(n - x + 2 - 2\sqrt{n-x}\,\right) = n + 2 - \sqrt{4(n-x)}$. The most x can be is $|M| \leq n/2$, whence the bound. $\square$

The bound is achieved for a graph that is the corona of the extremal graphs for the original bound. Note that in this example a maximum matching is removed.

For total domination one needs that the matching removal does not leave any isolates; one could restrict to such matchings, but we will instead only study the total domination question for graphs with smallest degree at least 2. Even then it can be that $\gamma_t(G-M)=n$; consider for example G being an even cycle and M a maximum matching. The bound for the independence number is trivial: we have $\alpha(G-M)\le n-1$, with the bound achieved by the star, and M being any matching thereof.

We next consider how the values $\psi(G-M)$ and $\psi(G)$ compare. It is straightforward to show that $\psi(G-M)\le 2\psi(G)$. For, if ψ is a domination parameter, the removal of the matching can leave at most $\psi(G)$ vertices undominated. If ψ is the independence number, then starting with $G-M$ and adding back in the matching can at most halve the size of a maximum independent set. It is easy to find examples that achieve this bound. For instance, K_n is an example for γ, i and α for $n\ge 2$; more examples are given in the discussion of cubic graphs.

Similarly, except for the independent domination number, it is immediate that $\psi(G-M)\ge\psi(G)$, and there are many examples where there exists a maximum matching whose removal leaves its value unchanged. For instance, $K_{m,m}$ is an example for γ and α for $m\ge 2$; more examples are given in the discussion of cubic graphs. In contrast it can happen that $i(G-M)$ is much smaller than $i(G)$. A simple example is the double star (shown in Figure 1), where the independent domination number decreases from $n/2$ to 2 on the removal of the (maximal) matching consisting of the central edge.

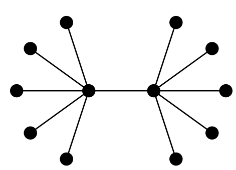

Figure 1: A double star

3 Trees and Bipartite Graphs

We consider next the removal of a matching in a tree.

The situation for domination was treated at length in [4]. We consider here the concept for independent domination first. As for domination, there are trees where the removal of a maximum matching can double the independent domination number. An example is the "double corona" of any tree; that is, taking a tree and adding two end-vertices to each vertex. For the minimum value of $i(T-M)$, we noted above that the double star shows that $i(T-M)$ can be much smaller than $i(T)$ (even if one requires the matching to be maximal). For domination, a tree always has a matching whose removal increases the domination number. Indeed we showed in [4] that this is true of any maximum matching. It turns out that the same is true of independent domination, though the proof is different:

Theorem 2 *If T is a nontrivial tree and M_m is a maximum matching of T, then $i(T-M_m) > i(T)$.*

Proof. The proof is by induction on the order of T. It is easy to check the result for trees of diameter 1 (K_2), diameter 2 (stars), and diameter 3 (double stars). So assume the diameter of T is at least 4.

Consider a path $v_0v_1v_2v_3\dots v_d$ of diametrical length in T. Think of the tree as rooted at v_d. By the choice of path, all children of v_1 are leaves, and all children of v_2 are either leaves or have only leaves as children.

Case 1: Suppose the edge $e = v_1v_2$ is in M. It follows that if w_1 is a child of v_2 then w_1 is not a leaf, else there is an M-augmenting path $v_0v_1v_2w_1$, a contradiction of the maximality of M. Let T' denote the tree formed by deleting from T the vertex v_1 and all its children. By the choice of diameter, T' is nontrivial. Any maximum matching of T' can be extended by the edge v_0v_1 to a matching of T. It follows that $M-e$ is a maximum matching of T'. See Figure 2(a).

We claim that $i(T) \le i(T')+1$. Consider an i-set J of T'. (As an anamnesis, an i-set of a graph is as per usual parlance an independent dominating set of minimum size.) If J does not contain v_2, then the bound follows, since one can extend J to an independent dominating set of T by adding v_1. So assume J contains v_2. Let p be a vertex of T' not dominated by $J-\{v_2\}$ (which exists by the minimality of J). Then p cannot be a child of v_2, since such a vertex has itself a child-leaf and that child-leaf must be in J. If p is v_3, then one can replace

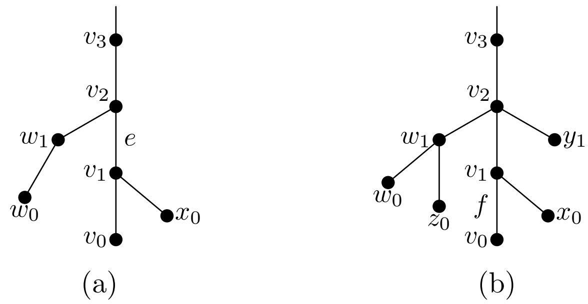

Figure 2: Two possible subtrees

v_2 in J by v_3 to get an i-set of T' that does not contain v_2. So we may assume the only choice for p is v_2 itself. But then one can add v_1 to J and remove v_2 to obtain an independent dominating set of T. The claim follows.

Now, apply the inductive hypothesis to obtain that $i(T' - (M - e)) > i(T')$. And $i(T - M) = i(T' - (M - e)) + 1$, since no other edge incident with v_1 is in M. That is,

$$i(T - M) = i(T' - (M - e)) + 1 > i(T') + 1 \geq i(T),$$

and this case is complete.

Case 2: Suppose that the edge $f = v_0v_1$ is in M. If v_1 has degree 2 in T, then as above let T' denote the tree formed by deleting from T the vertices v_1 and v_0. Similar to before, $M - f$ is a maximum matching of T' and $i(T) \leq i(T') + 1$. Further, $i(T - M) \geq i(T' - (M - f)) + 1$, since the vertex v_1 is a leaf in $T - M$ and hence its addition cannot decrease i, while v_0 is isolated in $T - M$ and so must be in every i-set. Applying the inductive hypothesis it holds that

$$i(T - M) \geq i(T' - (M - f)) + 1 > i(T') + 1 \geq i(T).$$

as required.

So we may assume, for all choices of v_1, that v_1 has at least two children. Say x_0 is another child of v_1. See Figure 2(b). There are three subcases. First, assume that v_2 has no child that is a leaf. Then define T' as before by deleting v_1 and its children. As in Case 1, one may argue that if v_2 is in every i-set J of T or of T', then $J - \{v_2\}$ dominates all vertices except v_2. So $i(T - M) \geq i(T' - (M - f)) + 1$

while $i(T) \leq i(T') + 1$, and the induction hypothesis yields the desired bound, as before.

Second, assume v_2 has a child y_1 that is a leaf where $v_2 y_1$ is not in M. Then let T'' be the tree formed from T by the removal of both x_0 and y_1. Note that M is a maximum matching of T''. We claim that $i(T) = i(T'') + 1$ and $i(T - M) = i(T'' - M) + 1$. For consider an i-set J of any one of the four graphs T, T'', $T - M$, or $T'' - M$. Then if v_2 is not in J, one may assume v_1 is in, since v_1 has one non-leaf neighbor; that is, one may assume exactly one of v_1 and v_2 is in J. It follows further that when J is coming from T or $T - M$, one may assume exactly one of x_0 and y_1 is in J, and the claim follows. So one can use the inductive hypothesis to conclude that $i(T - M) = i(T'' - M) + 1 > i(T'') + 1 = i(T)$, as required.

Third, assume that the only leaf child y_1 of v_2 is such that $y_1 v_2$ is in M. Let T^* be the tree formed from T by the removal of v_2 and all its descendants. Let M^* denote M restricted to T^*. Note that M^* is a maximum matching of T^*. Say v_2 has c children in total. Then $i(T) \leq i(T^*) + c$, since any independent dominating set of T^* can be expanded to one of T by incorporating all of v_2's children. Further, every independent dominating set of $T - M$ must contain y_1 as well as at least two vertices for each other child of v_2 (since in T such a child has at least two children and one of them is isolated in $T - M$). Thus $i(T - M) \geq i(T^* - M^*) + (2c - 1)$. By the induction hypothesis one can conclude that $i(T - M) \geq i(T^* - M^*) + (2c - 1) > i(T^*) + (2c - 1) > i(T)$, as required. $\square$

Theorem 2 is best possible in a couple of ways. First, as noted with the double star, it can happen that $i(T - M) < i(T)$ even if matching M is required to be maximal. Second, for all positive integers r, there exist trees T such that $i(T - M_m) \leq i(T) + 1 = r + 1$ for every maximum matching M_m of T. Example trees include those for domination described in [4] (a particular example being the wounded spider shown in Figure 3).

Lemma 1 showed that $i(G - M)$ can be close to n. Nevertheless, for trees and indeed for bipartite graphs there is a better bound. Recall that an isolate-free bipartite graph has independent domination number at most half its order (since both partite sets are independent dominating sets). We have the following bound, which coincides with the bound for ordinary domination for this class of graphs:

Theorem 3 *For a connected bipartite graph B with order $n \geq 3$ and matching M, it holds that $i(B - M) \leq 3n/4$ and this bound is sharp.*

PROOF. The graph $B - M$ is bipartite. Thus $i(B - M)$ is at most half the number of non-isolates plus the number of isolates. The number of isolates is at most $|M| \leq n/2$, whence the bound. The bound is achieved (only) by the corona of a bipartite graph B', where B' is a bipartite graph with $i(B')$ equal to half its order. $\square$

For a tree (and hence forest) of order n the independence number is at least $n/2$ and at most $n - 1$ (unless empty). It follows that $\alpha(T - M) \leq 2\alpha(T) - 1$ if T is a nontrivial tree and M is a matching. Further, there is a unique family of graphs with equality in this bound, namely the stars where all but one edge is subdivided once. An example is shown in Figure 3.

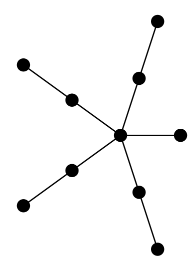

Figure 3: A wounded spider

4 Domination and Independent Domination in Cubic Graphs

We study cubic graphs next. In particular, we consider both the maximum and minimum values, and the maximum and minimum values of the ratios of the "before" and "after" value of the parameter.

For domination, the upper bound for a cubic graph Q and matching M is $\gamma(Q - M) \leq n/2$ since $Q - M$ is connected. An extremal example is the bracelet formed by taking disjoint diamonds (that is, copies of $K_4 - e$) and adding edges to form a connected graph. See Figure 4 for an example. In a bracelet there exists a perfect matching M whose removal leaves the disjoint union of 4-cycles. It is also true that $i(Q - M) \leq n/2$ for any cubic graph Q and matching M, since that is the bound for connected graphs with maximum degree at least 3, as shown in [1].

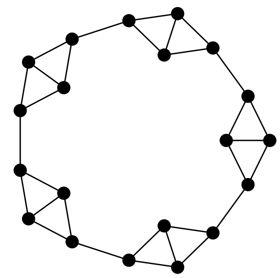

Figure 4: A bracelet

We consider lower bounds for $\psi(Q - M)$ next. The smallest $\gamma(Q)$ can be is $n/4$. It can happen that $\gamma(Q - M) = n/4$ even if one requires M to be maximal. Indeed, the above bracelet provides an example. If there is an even number of diamonds, then there is a maximal matching M_ℓ that misses one inner vertex from each diamond, and these uncovered vertices form an efficient/perfect dominating set of both Q and $Q - M_\ell$. The same result holds for independent domination.

Turning to maximum matchings, we note that if Q has a perfect matching M_p, its removal yields a 2-regular graph, and so $i(Q - M_p) \geq \gamma(Q - M_p) \geq n/3$. However, for maximum matchings in cubic graphs without perfect matchings the lower bound is smaller.

We will need to examine the structure of the maximum matching in a cubic graph. Recall that for a graph H the notation $o(H)$ denotes the number of components of graph H of odd order. And that the Tutte–Berge formula relates the matching number of a graph to the number of odd components in some subgraph. In particular:

Theorem 4 *[2] If M_m is a maximum matching of a graph Q, then there is a vertex set T such that the number of uncovered vertices is $o(Q - T) - |T|$.*

Furthermore, the result implies that each vertex of T is matched with a vertex of a different odd component of $Q - T$, while the uncovered vertices are one in each of the remaining odd components. This idea was used, for example, in the determination of the minimum size of a maximum matching of a cubic graph, as given by Biedl et al. [3]. Indeed, one can readily attain a bound by using their result, but it is not quite as good as the following:

Theorem 5 *If Q is a cubic graph of order n and M_m a maximum matching of Q, then $\gamma(Q - M_m) \geq 10n/33$.*

Proof. Let T be the subset such that the number of uncovered vertices equals $o(Q-T) - |T|$. Let $t = |T|$ and let d denote the count of the odd components of $Q - T$. By parity, there is an odd number of edges between an odd component and T.

Call an odd component of $Q - T$ "solo" if it has only one edge to T. Let s denote the number of solo components. Note that $s \leq d$. By considering the edges between the odd components and T, we have $s + 3(d - s) \leq 3t$ which simplifies to $3d - 2s \leq 3t$. Since Q is connected, if we contract each odd component to one vertex, and contract all even components (if any) to one combined vertex, we have a connected graph of order $d + t$ or $d + t + 1$ all of whose edges are incident with T. Thus $d + t \leq 3t + 1$ which simplifies to $d \leq 2t + 1$.

Call a solo component "free" if the edge joining it to T is in M_m. Let s_0 denote the number of solo components that are not free. Note that $s_0 \leq t$ and that $t \leq d - s_0$, since every vertex of T is matched with an odd component. Let s_5 and s_7 denote the number of free components of orders 5 and 7 respectively. Note that $s_0 + s_5 + s_7 \leq s$. Since each solo component has at least five vertices, we have $n \geq t + 5s_0 + 5s_5 + 7s_7 + 9(s - s_5 - s_7 - s_0) + 1(d - s)$, which simplifies to $n \geq t + 8s - 4s_0 - 4s_5 - 2s_7 + d$.

The number of uncovered vertices is $d-t$. So in $Q-M$ there are $d-t$ vertices that dominate four vertices, and the rest dominate three. Now in $Q - M_m$, each free component of order 5 actually needs 2 vertices to dominate it, and every free component of order 7 actually needs 3 vertices to dominate it. Hence $3\gamma + (d - t) \geq n + s_5 + 2s_7$, where γ is the domination number of $Q - M_m$.

Then we have linear program

$$
\begin{aligned}
&\text{minimize } (n + s_5 + 2s_7 - (d - t))/3 \\
&\text{subject to } s \le d \\
&\qquad 3d - 2s \le 3t \\
&\qquad d \le 2t + 1 \\
&\qquad s_0 \le t \\
&\qquad t \le d - s_0 \\
&\qquad s_0 + s_5 + s_7 \le s \\
&\qquad n \ge t + 8s - 4s_0 - 4s_5 - 2s_7 + d
\end{aligned}
$$

Dividing through by n (and ignoring the $+1$ on $2t + 1$), this LP is minimized with a value of $10/33 \approx 0.303$ where $s_0 = 1/11$, $s_5 = s_7 = 0$, $s = 3/22$, $d = 2/11$ and $t = 1/11$. $\square$

One can come close to building such a graph. The actual bound is not achievable, since such a graph would after matching-deletion have a component of order 13 (two s_0-pieces joined by a path that goes T, singleton component in $Q - T$, then T again), and $13 - 8$ is not a multiple of 3, so the component cannot be efficiently dominated using the two vertices of degree 3. If we change the s_0 components to have 7 vertices, consider the following 26-vertex fragment shown in Figure 5. Let Q_k denote the connected cubic graph formed by taking $k \ge 2$ copies of the fragment and joining up. We claim that the red edges form a maximum matching M_m. (Note that the ringed vertices form a set T of $2k$ vertices whose removal yields $4k$ odd components, and so leaving $2k$ vertices uncovered is optimal.) Further, the fragment after removal of the matching has a dominating set of size 8. So Q_k has order $26k$ and $Q_k - M_m$ has domination number $8k$. And $4/13 \approx 0.308$.

We turn now to the ratio $\psi(G - M)/\psi(G)$. As noted earlier, the ratio can be as large as 2. This is achieved by the bracelets (see Figure 4). And for ordinary domination, the ratio is always at least 1; again equality is achieved by the bracelets.

However, as observed above, removing a matching can decrease the independent domination number of a graph. Here for example, removing any (nonempty) matching from $K_{3,3}$ reduces its independent domination number from 3 to 2.

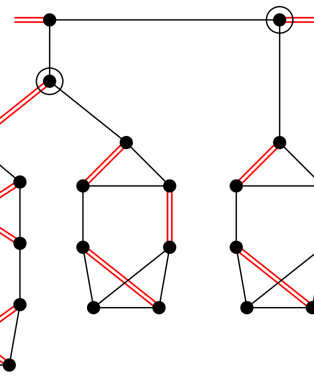

Figure 5: Part of a cubic graph and its maximum matching

What about other cubic graphs? Except for $K_{3,3}$ we have the chain:

$$i(G-M) \geq \gamma(G-M) \geq \gamma(G) \geq \frac{3}{4}i(G).$$

(The last inequality is a theorem from [9].)

Two families of cubic graphs with $i = 3n/8$ are described in [9]. All these graphs are hamiltonian; and so removing the perfect matching M_p of the edges not in the hamiltonian cycle leaves a graph with $i(G-M_p) = \lceil n/3 \rceil$. This example gives an asymptotically $8/9$ ratio for $i(G-M_p)/i(G)$. This would be the smallest possible for cubic graphs with perfect matchings if the conjecture from [9] is true; namely that $K_{3,3}$ and the 5-prism (shown in Figure 6) are the only cubic graphs with independent domination number more than $\frac{3}{8}$ their order.

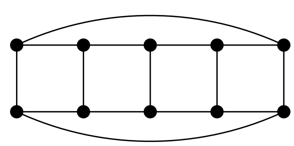

Figure 6: The 5-prism

5 Independence in Cubic Graphs

For independence number the lower bound is trivial: apart from K_4 all cubic graphs Q are 3-colorable and therefore have $\alpha(Q) \geq n/3$. Further, if Q has a spanning subgraph consisting of the disjoint union of triangles, then the other edges form a perfect matching M_p such that $\alpha(Q - M_p) = n/3$.

For an upper bound, we note that the graph $Q - M$ has maximum degree at most 3 and minimum degree at most 2. Thus if J is a maximum independent set thereof, the number of edges incident with J is at least $2|J|$ and at most $3(n-|J|)$ so that $\alpha(Q - M) \leq \frac{3}{5}n$. This value can be achieved where $Q - M$ is the disjoint union of $K_{2,3}$. See Figure 7 for an example.

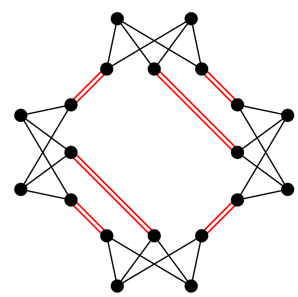

Figure 7: A cubic graph with $4K_{2,3}$ as spanning subgraph

On the other hand, it is immediate that $\alpha(Q - M_m) \leq n/2$ for any perfect matching M_m, and this bound can be attained (for example by the bracelets again). We next show that removing a maximum matching from a cubic graph always yields a graph with independence number at most half its order. We will have cause to avail ourselves of the following ceiling on the independence number of a "nearly" 2-regular graph:

Lemma 6 *Let H be a graph with minimum degree at least 2 and maximum degree at most 3, with ℓ vertices of degree 3. Then $\alpha(H) \leq n/2 + \ell/4$.*

Proof. The proof is by induction on ℓ. We know the bound is true for $\ell = 0$ (that is, when H is the union of cycles). So assume $\ell > 0$. Necessarily ℓ is even; so H has at least two vertices of degree 3. Consider a shortest path P from a vertex of degree 3 to another vertex of degree 3. The interior vertices of P have degree 2; say there are k of them (where possibly $k = 0$). Remove the interior vertices of P (or the single edge if it has length 1) to form graph H'. By the inductive hypothesis, the graph H' has independence number at most

$(n-k)/2 + (\ell - 2)/4$. Adding back in the k interior vertices of P can increase the independence number by at most $(k+1)/2$. The bound follows. $\square$

Lemma 7 *If Q is a cubic graph with maximum matching M_m leaving ℓ vertices uncovered, then*

$$o(Q - M_m) \geq \ell.$$

Proof. Let T be the optimal set from the Tutte–Berge formula (Theorem 4); that is, $o(Q - T) = \ell$. Assign the vertex set of each odd component of $Q - T$ to a "box". At the start there are $|T| + \ell$ boxes. The remaining vertices are labeled surplus. Note that in $Q - T$ there is no edge between boxes nor with exactly one end in a box. And similarly with $Q - T - M_m^*$ where M_m^* is the edges of M_m not incident with T.

Consider building $Q - M_m$ from $Q - T - M_m^*$ as follows. Consider some edge of $Q - M_m$ with exactly one end in a box (and necessarily other end in T). Since every vertex outside a box has degree 2 in $Q - M_m$, this edge can be extended to a path in $Q - M_m$ whose other end is a vertex in a box (possibly the same box).

If the two ends of the path are in the same box, then move (the contents of) the box to surplus. Otherwise, if the path has length 2, then its unique interior vertex is in T, and merge the two boxes and add the vertex of T to the resultant box. If the path has length more than 2, then move (the contents of) the two boxes containing the two ends to surplus. Note that in this case the path contains at least two vertices of T. The key fact is that each step and hence the entire process reduces the number of boxes by (at most) the number of vertices of T. And so at the end one is left with at least ℓ live boxes.

Now, consider a remaining box. In $Q - M_m$ there is no edge with exactly one end in the box. Further, note that the box has an odd number of vertices: the merge process combines two odd sets with one vertex of T. That is, there is an odd component of $Q - M_m$ contained in the box. The lemma follows. $\square$

Theorem 8 *If Q is a cubic graph of order n and M_m a maximum matching of Q, then $\alpha(Q - M_m) \leq n/2$.*

Proof. Say the components of $Q - M_m$ are $H_1, \ldots, H_k$ with orders $n_1, \ldots, n_k$ and $s_1, \ldots, s_k$ vertices of degree 3 respectively. Note that each s_j is even, and that they sum to ℓ, the number of uncovered vertices. By Lemma 6 it holds that $\alpha(H_j) \leq n_j/2 + s_j/4$. Note that this expression is half-integral; thus if the bound is not achieved then $\alpha(H_j) \leq n_j/2 + s_j/4 - \frac{1}{2}$. By Lemma 7 there are at least ℓ components where n_j is odd. For these components, the lower bound cannot hold with $s_j = 0$. And there are at most $\ell/2$ components with $s_j > 0$. Thus

$$\alpha(G - M_m) = \sum_j \alpha(H_j) \leq \left(\sum_j n_j/2\right) + \left(\sum_j s_j/4\right) - (\ell/2)(\frac{1}{2}) = n/2,$$

whence the result. $\square$

Note that equality in Theorem 8 is possible even if M_m is not a perfect matching. An example is shown in Figure 8; in this depiction the hollow vertices constitute a maximum independent set of $Q - M_m$.

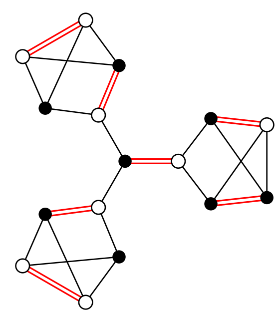

Figure 8: The smallest cubic graph without a perfect matching

Consider next the ratio question. We observed above that the minimum value of $\alpha(Q)$ is $n/3$ (apart from K_4) while the maximum value of $\alpha(Q - M_m)$ where M_m is a maximum matching is $n/2$ by the above theorem. So $\alpha(Q - M_m)/\alpha(Q) \leq 3/2$ for $n \geq 6$, and this is achievable, for example, by the inflation of a bipartite cubic graph, such as shown in Figure 9. (Recall that the inflation of a cubic graph H is the line graph of the subdivision of H.)

However, there are graphs where the ratio $\alpha(Q - M)/\alpha(Q)$ is larger. For example, let F be the 18-vertex graph shown in Figure 10. If one takes multiple

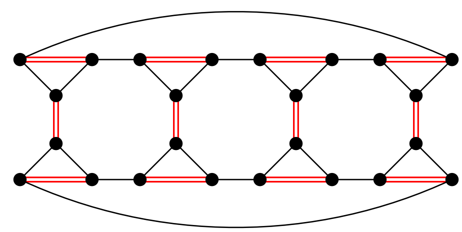

Figure 9: The inflation of the cube

copies of F and adds edges to make it a connected cubic graph, the result has ratio $\frac{5}{3}$. This is best possible.

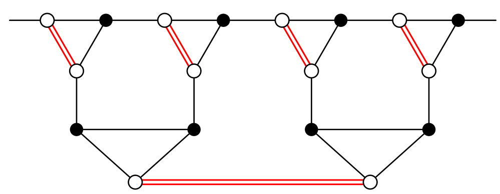

Figure 10: A fragment of a cubic graph

Theorem 9 *If Q is a connected cubic graph other than K_4 and M any matching of Q, then $\alpha(Q-M)/\alpha(Q) \leq \frac{5}{3}$.*

Proof. Let J be a maximum independent set in $Q-M$ with size j. Let b denote the number of edges between J and $V-J$ in $Q-M$. Since every vertex in $Q-M$ has degree at least 2, it holds that $b \geq 2j$. Say $V-J$ has ℓ vertices with degree 3 to J, and s vertices with degree 2 to J. Then $b \leq 3\ell + 2s + (n-j-\ell-s)$. So we have the inequalities $2j \leq 3\ell + 2s + (n-j-\ell-s)$ and $\ell + s \leq n-j$.

There are three lower bounds on the independence number of Q. By reintroducing M back into J it follows that $\alpha(Q) \geq j/2$; by reintroducing M back into $V-J$ it follows that $\alpha(Q) \geq \ell + s/2$; and since Q is 3-colorable it follows that $\alpha(Q) \geq n/3$.

Consider the problem of maximizing j/α subject to these constraints. If we divide the constraints by α, the result is a linear program (in variables j/α, ℓ/α,

s/α, and n/α). Software says that the maximum is $\frac{5}{3}$, achieved with $j = \frac{5}{9}n$, $\alpha = \frac{1}{3}n$, and variable ℓ, s including for example $\ell = s = \frac{2}{9}n$. $\square$

6 Total Domination in Cubic Graphs

For a cubic graph Q and matching M, the graph $Q - M$ has minimum degree 2. Therefore by the upper bound of [5], its total domination number is at most $2n/3$. This value is achievable by $Q - M$ being, for example, a disjoint union of triangles and 6-cycles.

Consider lower bounds on $\gamma_t(Q - M)$. It is immediate that $\gamma_t(Q - M_p) \geq n/2$ for a perfect matching, since the resultant graph is 2-regular. It can be shown that $\gamma_t(Q - M_\ell) \geq 2n/5$ for any maximal matching M_ℓ, and this is sharp where $Q - M_\ell$ is the disjoint union of $K_{2,3}$ (as in Figure 7). We next resolve the question where one is required to remove a maximum matching. We will need the the following lemma:

Lemma 10 *Let H be a graph with minimum degree at least* 2 *and maximum degree at most* 3*, with ℓ vertices of degree* 3*. Then $\gamma_t(H) \geq n/2 - \ell/2$.*

Proof. Let D be a total dominating set of H. Then at most ℓ vertices of D have 3 neighbors while the remainder have at most 2 neighbors. Since every vertex has a neighbor in D, it follows that $3(\ell) + 2(|D| - \ell) \geq n$. Rearranged this gives the desired bound. $\square$

Theorem 11 *If Q is a cubic graph of order n and M_m a maximum matching of Q, then $\gamma_t(Q - M_m) \geq n/2$.*

Proof. Say the components of $Q - M_m$ are $H_1, \ldots, H_k$ with orders $n_1, \ldots, n_k$ and $s_1, \ldots, s_k$ vertices of degree 3 respectively. Note that each s_j is even, and that they sum to ℓ, the number of uncovered vertices. By the above lemma, it holds that $\gamma_t(H_j) \geq (n_j - s_j)/2$. Note that equality can occur only if n_j is even (and when it does not $\gamma_t(H_j) \geq (n_j - s_j)/2 + \frac{1}{2}$); further, by Lemma 7 there are at least ℓ components where n_j is odd. Thus

$$\gamma_t(G - M_m) = \sum_j \gamma_t(H_i) \geq \left(\sum_j n_j/2\right) - \left(\sum_j s_j/2\right) + \ell(\frac{1}{2}) = n/2,$$

whence the result. $\square$

This bound can be achieved even if M_m is not perfect matching; consider for example the smallest cubic graph without a perfect matching.

For the ratio question, the removal of a matching can double the total domination number. This ratio is achievable. Consider for example the inflation $I(H)$ of a cubic graph H where H has a perfect matching X. The inflation has total domination number $\frac{1}{3}$ its order, by taking both ends of every edge of $I(H)$ corresponding to X. At the same time, if M is the matching of $I(H)$ consisting of all edges not in a triangle, then $I(H) - M$ consists of disjoint triangles and so has total domination number $\frac{2}{3}$ its order.

On the other hand, there are cubic graphs where the removal of a perfect matching does not change the total domination number. For example, consider the cubic graphs with total domination number equal to half their order, as determined in [11]. All these graphs have order n a multiple of 4; further, they are all hamiltonian, and so choosing a hamilton cycle and removing the edges not in that cycle leaves a graph with total domination number $n/2$.

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